Sorting_weight_Udemy / PLC_1 [CPU 1212C AC/DC/Rly] / Program blocks

Block_1 [FC1]

Block_1 Properties							
General							
Name	Block_1	**Number**	1	**Type**	FC	**Language**	SCL
Numbering	automatic						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Block_1

Name	Data type	Default value	Comment
▼ Input			
recipe	Int		
count	Int		present counts of counter
weight	Real		sensor
at_scale	Bool		on the sensor
▼ Output			
left roller	Bool		
right roller	Bool		
weight_display	DInt		
▼ temp	Array[0..2] of Real		
temp[0]	Real		
temp[1]	Real		
temp[2]	Real		
InOut			
▼ Temp			
▼ Sorting	Array[0..5] of Bool		
Sorting[0]	Bool		
Sorting[1]	Bool		
Sorting[2]	Bool		
Sorting[3]	Bool		
Sorting[4]	Bool		
Sorting[5]	Bool		
Constant			
▼ Return			
Block_1	Void		

```
0001 //Recipe Logic
0002 IF #recipe = 0 THEN
0003   #Sorting[0] := 0;
0004   #Sorting[1] := 0;
0005   #Sorting[2] := 0;
0006   #Sorting[3] := 1;
0007   #Sorting[4] := 1;
0008   #Sorting[5] := 0;
0009 END_IF;
0010
0011 IF #recipe = 1 THEN
0012   #Sorting[0] := 0;
0013   #Sorting[1] := 1;
0014   #Sorting[2] := 1;
0015   #Sorting[3] := 0;
0016   #Sorting[4] := 0;
0017   #Sorting[5] := 0;
0018 END_IF;
0019
0020 IF #recipe = 2 THEN
0021   #Sorting[0] := 1;
0022   #Sorting[1] := 0;
0023   #Sorting[2] := 0;
0024   #Sorting[3] := 0;
0025   #Sorting[4] := 0;
0026   #Sorting[5] := 1;
0027 END_IF;
0028
0029 IF #recipe > 2 OR #recipe < 0 THEN
0030   #recipe := 0;
0031 END_IF;
0032
0033 //Storing Box information
0034 IF #at_scale = 1 THEN
0035   #temp[#count] := #weight;
0036 END_IF;
0037
0038 //Large Box
0039 IF #temp[#count] > 7.0 AND #temp[#count] < 8.0 THEN
0040   #"left roller" := #Sorting[0];
0041   #"right roller" := #Sorting[1];
0042 END_IF;
0043
0044 //Medium Box
0045 IF #temp[#count] > 4.5 AND #temp[#count] < 5.5 THEN
```

```
0046   #"left roller" := #Sorting[2];
0047   #"right roller" := #Sorting[3];
0048 END_IF;
0049
0050 //Small Box
0051 IF #temp[#count] > 3.5 AND #temp[#count] < 4.5 THEN
0052   #"left roller" := #Sorting[4];
0053   #"right roller" := #Sorting[5];
0054 END_IF;
0055 //For displaying Weight on Panel
0056 #weight_display := REAL_TO_DINT(#weight * 10.0) ;
0057
0058
```

Symbol	Address	Type	Comment
#"left roller"		Bool	
#"right roller"		Bool	
#at_scale		Bool	on the sensor
#count		Int	present counts of counter
#recipe		Int	
#Sorting		Array	
#temp		Array	
#weight		Real	sensor
#weight_display		DInt	